Ryan Hartung

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Education

The University of Notre Dame Notre Dame, IN
Ph.D. in Computer Science & Engineering *Expected May 2029*

- Advisor: Dr. Douglas Thain
- Research Focus: Distributed Systems and High-Performance Computing

The University of Notre Dame Notre Dame, IN
Master of Science in Computer Science & Engineering *August 2025*

The University of Texas at Austin Austin, TX
Bachelor of Science in Physics (Computation Option) *May 2024*

- Certificate in Elements of Computing

Publications

[2026-5] Liubov Kurafeeva, **Ryan Hartung**, Benjamin Carter, Alan Subedi, Avhishek Biswas, Michael Fay, Shantenu Jha, Chandra Krintz, Andre Merzky, Douglas Thain, Memet Can Vuran, and Rich Wolski. "Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference." *arXiv preprint arXiv:2605.20532*. https://arxiv.org/abs/2605.20532

[2025-11] Liubov Kurafeeva, Alan Subedi, **Ryan Hartung**, Michael Fay, Avhishek Biswas, Shantenu Jha, Ozgur Kilic, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture." In *Proceedings of the SC '25 Workshops*. ACM, 2317–2327. DOI: https://doi.org/10.1145/3731599.3767589

Research Experience

The University of Notre Dame Notre Dame, IN
Graduate Research Assistant January 2025 – Present

- Contributing to the design and implementation of xGFabric, architectural framework connecting remote sensor edge networks to HPC centers via 5G.
- Developing low-latency data pipelines to facilitate real-time inference across heterogeneous edge-HPC infrastructure.

The University of Texas at Austin Austin, TX
Undergraduate Research Assistant June 2022 – December 2023

- Engineered an open-source Python tool analyzing star-spot patterns across 6,000+ star systems with optimized array processing algorithms.
- Evaluated star-spot contrast ratios using observational data from NASA Kepler and TESS missions to establish baseline metrics for exoplanet detection.
- Integrated parallel processing and multi-core compilation to optimize data throughput and execution runtime.
- Published codebase and workflows to an open-source GitHub repository; research funded by NASA grant.

Work Experience

Hargrove Engineers and Constructors Katy, TX
Controls & Automations Engineering Intern May 2024 - August 2024

- Automated the migration of 200,000+ lines of Dow MOD5 code into Honeywell Experion without the loss of functionality
- Tracked, analyzed, and documented over 7,000 variables to design and draw comprehensive flow diagrams to visualize variable definitions
- Developed multiple algorithms to expedite the migration and variable mapping processes, improving overall efficiency

Hargrove Engineers and Constructors Katy, TX
Controls & Automations Engineering Intern May 2023 - August 2023

- Instantiated existing library modules to complete the upcoming Reactor and Train configuration for Westlake's PVC plant
- Performed safety testing on all module and sequence configurations using internal procedures on simulated ABB hardware
- Successfully configured Modbus communication between the ABB and HIMA system controllers
- Updated structured text configuration to handle cross-controller communications

Talks, Panels & Presentations

XLOOP Workshop at SC '25 — St. Louis, MO
Invited Panelist — November 16 2025

- Participated in expert panel on edge-to-HPC loop architectures [Recording available online].

GCASR 2026 — Evanston, IL
Poster Presenter — May 11, 2026

- Poster: *Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference*.

HPDC 2025 — Notre Dame, IN
Poster Presenter — July 21, 2025

- Poster: *xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture*.

Teaching Experience

The University of Notre Dame — Notre Dame, IN
Graduate Teaching Assistant — August 2024 – May 2025

- Elements of Computing II — *Spring 2025*
- Principles of Computing — *Fall 2024*

Technical Skills

- **Programming Languages:** C/C++, Python, Rust, Java, JavaScript, Julia, HTML, JSON, XML
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, Keras
- **Distributed Systems & HPC:** SLURM, HTCondor, Univa Grid Engine (UGE)
- **Tools, Platforms & Automation:** ABB, Git, Honeywell, LaTeX, Linux/UNIX, Microsoft Suite, OpenFOAM, ParaView

Leadership & Service

Order of Christian Initiation of Adults (OCIA), Sponsor — September 2026 – Present
Fischer Graduate Residence Mass, Lector — August 2025 – May 2026
Society of Catholic Scientists, Member — August 2025 – Present
Science Policy Initiative at Notre Dame, Member — August 2025 – Present
Catholic Graduate Community, Member — April 2025 – Present
Technology and Catholicism Club, Member — February 2025 – Present
Notre Dame Rock Climbing Club, Member — August 2024 – Present
Notre Dame Texas Club, Member — August 2024 – Present
Science Olympiad, Competitor, Mentor, Club President, Proctor Event-Supervisor, and Lab Coordinator) — August 2011 – Present